

## Hoofdstuk 3

## Lesing 3.3

①

\* KCL  $\rightarrow$  Node analyse (Supernodesse)

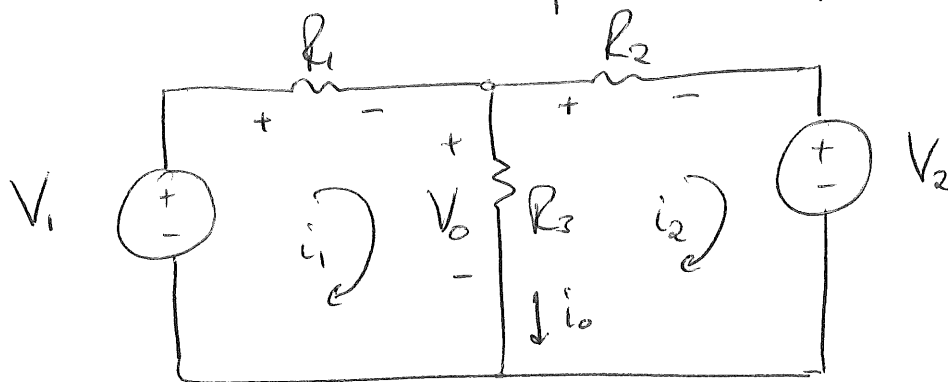
\* KVL  $\rightarrow$  Lns analyse / "Mesh analysis"

\* Skryf spannings its strome en weerstande

Stroomlus:  $\rightarrow$  onafhanklike lns.

(Mesh current  $\Rightarrow$  independent loop.)

$V_0$



①  $i_1$  en  $i_2$  is lusstrome.

$i_0$  is nie 'n lusstroom nie.

$$\underbrace{i_1}_{\leftarrow} \quad \downarrow i_0 \quad \leftarrow i_2$$

$$\therefore \text{KCL: } i_0 + i_2 = i_1$$
$$\underline{i_0 = i_1 - i_2} \rightarrow$$

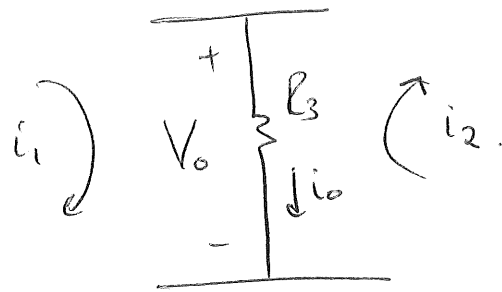
② KVL in elke onafh lns.

$$\Rightarrow -V_1 + i_1 R_1 + V_0 = 0$$

$$\Rightarrow -V_0 + i_2 R_2 + V_2 = 0.$$

(2)

✱



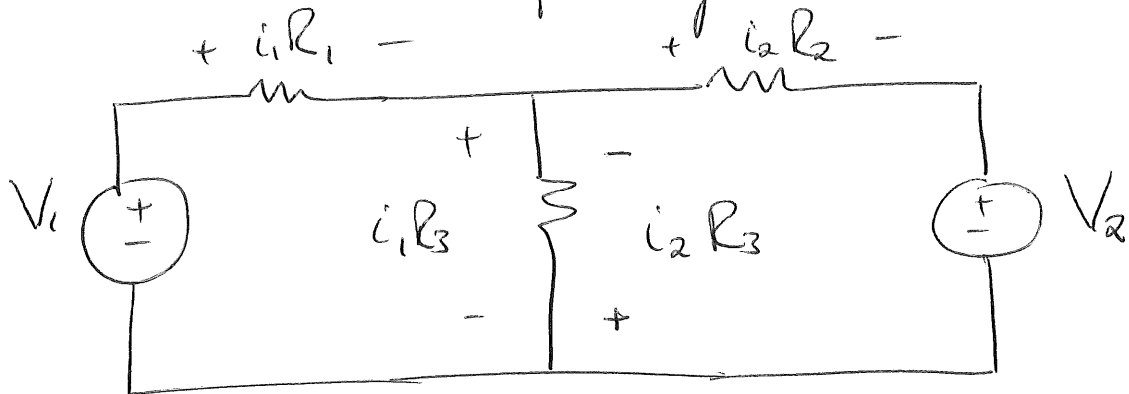
$$i_0 = i_1 - i_2.$$

$$* V_0 = i_0 R_3 = (i_1 - i_2) R_3$$

$$= i_1 R_3 - i_2 R_3$$

\* Spanning van  $b_0$  na onder.

\* Spanning van onder na  $b_0$ .



$$\text{Lus 1} \rightarrow -V_1 + i_1 R_1 + i_1 R_3 - i_2 R_3 = 0$$

$$i_1 (R_1 + R_3) - i_2 R_3 = V_1$$

$$\text{Lus 2} \rightarrow -i_1 R_3 + i_2 R_3 + i_2 R_2 + V_2 = 0$$

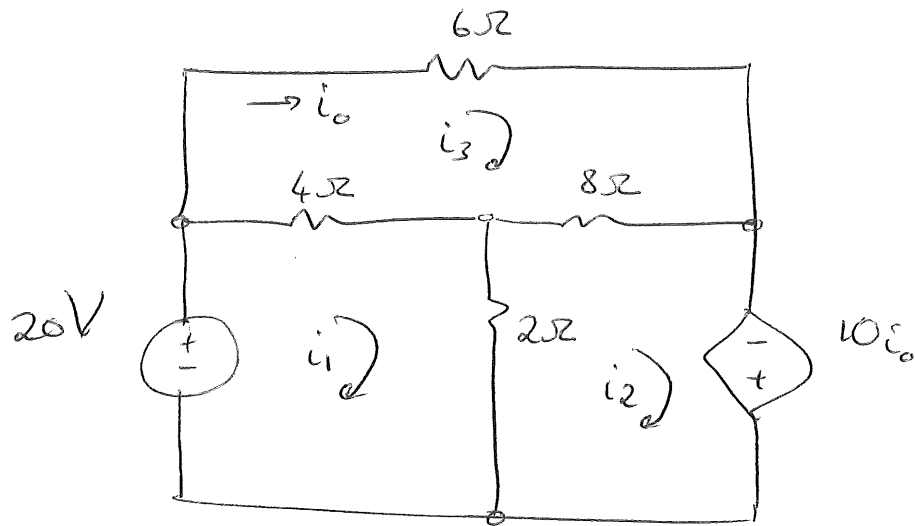
$$-i_1 R_3 + i_2 (R_2 + R_3) = -V_2.$$

$R$  wat door lus 2  
niet in onder lus  
gedeel word.

Som van  $R$ 'e in  
lus 2.

1/6

(3)



\* Let op:  $i_0 = i_3$

Loop 1:  $-20 + i_1(4 + 2) - 2i_2 - 4i_3 = 0$   
 $6i_1 - 2i_2 - 4i_3 = 20$

Loop 2:  $+i_2(2 + 8) - 2i_1 - 8i_3 - 10i_0 = 0$   
 $-2i_1 + 10i_2 - 8i_3 - 10i_3 = 0$   
 $-2i_1 + 10i_2 - 18i_3 = 0$

Loop 3:  $+i_3(4 + 6 + 8) - 4i_1 - 8i_2 = 0$   
 $-4i_1 - 8i_2 + 18i_3 = 0$

$$\begin{bmatrix} 6 & -2 & -4 \\ -2 & 10 & -18 \\ -4 & -8 & 18 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 0 \end{bmatrix}$$

V6

